

October 28, 2025

1. lavaan() SEM example using Wheaton et al. Data

```
> model <- ' # CRITICAL: NOTE SINGLE QUOTES
+ # measurement model
+   ses66 =~ educ66 + occstat66
+   alien67 =~ anomia67 + pwless67
+   alien71 =~ anomia71 + pwless71
+ # structural model
+   # direct effect
+     alien71 ~ c*ses66
+   # mediator
+     alien67 ~ a*ses66
+     alien71 ~ b*alien67
+   # indirect effect (a*b)
+     ab := a*b
+   # total effect
+     total := c + (a*b)
+ ' # END SINGLE QUOTES

> fit <- sem(model, data=wheaton) # fit model

> summary(fit) # examine fit

lavaan (0.5-22) converged normally after 59 iterations

Number of observations                                     932

Estimator                                                  ML
```

Minimum Function Test Statistic	76.184
Degrees of freedom	6
P-value (Chi-square)	0.000

Parameter Estimates:

Information	Expected
Standard Errors	Standard

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)
ses66 =~				
educ66	1.000			
occstat66	0.592	0.043	13.699	0.000
alien67 =~				
anomia67	1.000			
pwless67	0.823	0.038	21.735	0.000
alien71 =~				
anomia71	1.000			
pwless71	0.825	0.039	21.307	0.000

Regressions:

		Estimate	Std.Err	z-value	P(> z)
alien71 ~					
ses66	(c)	-0.172	0.064	-2.689	0.007
alien67 ~					
ses66	(a)	-0.759	0.062	-12.239	0.000
alien71 ~					
alien67	(b)	0.710	0.056	12.607	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)
.educ66	3.679	0.416	8.843	0.000
.occstat66	2.345	0.172	13.650	0.000
.anomia67	4.301	0.364	11.808	0.000
.pwless67	3.422	0.260	13.150	0.000
.anomia71	3.636	0.369	9.848	0.000
.pwless71	3.951	0.289	13.681	0.000
ses66	5.857	0.557	10.517	0.000
.alien67	5.200	0.495	10.514	0.000
.alien71	3.352	0.382	8.782	0.000

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)
ab	-0.539	0.057	-9.381	0.000
total	-0.711	0.061	-11.641	0.000

2. MGA Example in R

Unconstrained Model

lavaan (0.5-22) converged normally after 42 iterations

Number of observations per group

1	793
0	881

Estimator ML

Minimum Function Test Statistic 0.000

Degrees of freedom 0

Minimum Function Value 0.000000000000000

Chi-square for each group:

1	0.000
0	0.000

Parameter Estimates:

Information	Expected
Standard Errors	Standard

Group 1 [1]:

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)
trstlead =~				
trstplt	1.000			
trstprrt	0.979	0.031	31.657	0.000
trstprrl	0.707	0.033	21.352	0.000

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	3.916	0.076	51.203	0.000
.trstprrt	3.845	0.077	50.235	0.000
.trstprrl	4.649	0.077	60.609	0.000
trstlead	0.000			

Variances:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	0.591	0.104	5.707	0.000
.trstprrt	0.764	0.103	7.444	0.000
.trstprrl	2.644	0.142	18.653	0.000
trstlead	4.046	0.251	16.094	0.000

Group 2 [0]:

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)
trstlead =~				
trstplt	1.000			
trstprt	0.959	0.027	34.891	0.000
trstprl	0.837	0.031	26.904	0.000

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	3.831	0.074	51.735	0.000
.trstprt	3.879	0.073	53.017	0.000
.trstprl	4.289	0.078	55.268	0.000
trstlead	0.000			

Variances:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	0.677	0.090	7.481	0.000
.trstprt	0.894	0.089	10.081	0.000
.trstprl	2.396	0.129	18.630	0.000
trstlead	4.154	0.243	17.089	0.000

Note: A saturated model for comparison purposes...

Model with Loadings Constrained to be Equal

lavaan (0.5-22) converged normally after 40 iterations

Number of observations per group

1	793
0	881

Estimator	ML
Minimum Function Test Statistic	10.846
Degrees of freedom	2
P-value (Chi-square)	0.004

Chi-square for each group:

1	5.861
0	4.985

Parameter Estimates:

Information	Expected
Standard Errors	Standard

Group 1 [1]:

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)
trstlead =~				
trstplt	1.000			
trstprrt (.p2.)	0.968	0.020	47.318	0.000
trstprl (.p3.)	0.777	0.023	34.240	0.000

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	3.916	0.076	51.351	0.000
.trstprrt	3.845	0.076	50.685	0.000
.trstprl	4.649	0.080	58.199	0.000
trstlead	0.000			

Variances:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	0.584	0.085	6.904	0.000
.trstprrt	0.788	0.084	9.337	0.000
.trstprl	2.628	0.143	18.437	0.000
trstlead	4.027	0.233	17.280	0.000

Group 2 [0]:

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)
trstlead =~				
trstplt	1.000			
trstprrt (.p2.)	0.968	0.020	47.318	0.000
trstprl (.p3.)	0.777	0.023	34.240	0.000

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	3.831	0.074	51.586	0.000
.trstprrt	3.879	0.074	52.560	0.000
.trstprl	4.289	0.075	57.076	0.000
trstlead	0.000			

Variances:

	Estimate	Std.Err	z-value	P(> z)
.trstplt	0.676	0.086	7.868	0.000
.trstprrt	0.876	0.086	10.224	0.000
.trstprl	2.449	0.128	19.107	0.000
trstlead	4.183	0.233	17.969	0.000

Comparing the Chi-square statistic for this constrained model to the saturated model (with two parameters differing between the models), we would reject the null hypothesis that the two groups have equal loadings! We do not have measurement invariance, even in the “weak” sense. The correlation appears to be weaker in the group with response rates higher than 50%! Perhaps this group struggles with measurement (gets through interviews quicker?).